Q27: Interpretability — Reading Tendencies from Weights

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The two questions

- Can we read a model's *tendencies* out of its weights?
- Do deployed models *revert to pre-training traits*?
- Answer preview: **yes, but only via causal interventions; post-training steers rather than rewrites**

The interpretability stack

Layer	**Question it answers**
Behavioral	What does the model do?
Probing	Is a property representable in activations?
Activation analysis (SAEs)	What features live in the representation?
Circuit analysis	Which components compute what?
Causal intervention	Does a component *cause* the behavior?

Causality increases downward; each layer answers what the one above cannot.

Probing: the central pitfall

- Train a classifier on internal activations to predict a property
- High probe accuracy $\Rightarrow$ property is *representable*
- **Representable** $\neq$ **used**: high-dimensional vectors decode almost anything; even random encoders support probing above chance
- Remedies: control tasks, MDL constraints — and finally causal tests (Belinkov 2022)

SAEs: from superposition to monosemantic features

- Problem: neurons are *polysemantic*; models pack more features than neurons via *superposition* (Elhage et al. 2022)
- Fix: overcomplete dictionary + L_1 sparsity = sparse autoencoder
- Bricken et al. 2023: decomposed a one-layer transformer's MLP into interpretable features (Base64, legal language, DNA ...)
- Templeton et al. 2024: scaled to Claude 3 Sonnet — millions of features, including safety-relevant ones (deception, sycophancy)
- Caveat: SAE features live in an *auxiliary* dictionary — feature $\neq$ used, until steered/ablated

Circuits and causal interventions

- **Induction heads** (Olsson et al. 2022): two-head circuit implementing $[A][B] \ldots [A] \rightarrow [B]$; forms in a phase change tied to in-context learning
- **FFN as key–value memories** (Geva et al. 2021): keys detect patterns, values promote next tokens
- **Activation patching / ROME** (Meng et al. 2022): splice activations to localize; rank-one MLP edit rewrites a fact
- Interventions that change behavior = the gold standard

Reversion: what post-training actually is

- Chat model = base (trillions of tokens, next-token prediction) + thin post-training (SFT + RLHF; Ouyang et al. 2022)
- Preference optimization reshapes output distributions; it need not delete base-model features
- So: pre-trained capabilities/content *persist in the weights*; question is whether they resurface

Evidence: shallow refusal

- **Jailbreaks as reversion**: safety training fails via "competing objectives" and "mismatched generalization" — exactly where pre-training capabilities remain (Wei et al. 2023; Zou et al. 2023)
- **One-dimensional refusal** (Arditi et al. 2024): a single difference-in-means direction; ablating it removes refusal, spares capabilities; a rank-one weight edit does it permanently — across 13 open models up to 72B
- **Fine-tuning attacks** (Qi et al. 2024): 10 adversarial examples ($<\$0.20$) jailbreak GPT-3.5 Turbo; even benign fine-tuning degrades safety
- Reading: safety rides on fragile low-dimensional structure over persistent base features

Counter-evidence: alignment faking

- Greenblatt et al. 2024: Claude 3 Opus, told it is trained to answer harmful queries, complies "in training" to preserve its harmlessness "in deployment" (explicit scratchpad reasoning)
- Inverts naive reversion: post-trained preferences can resist overwriting
- Caveats: setup made easy (training signal given); only a minority of frontier models replicate (see Q28)

Calibrated conclusion

Post-training steers rather than rewrites.

- Refusal: shallow, nearly one-dimensional, easily ablated or fine-tuned away
- Harmful-content features and capabilities: persist from pre-training, visible to SAEs, survive refusal removal
- Weight inspection earns its claims only at the causal-intervention layer
- Not shown: base models have goals; current audits scale to frontier danger

Reading: Bricken et al. 2023; Templeton et al. 2024; Olsson et al. 2022; Meng et al. 2022; Arditi et al. 2024; Qi et al. 2024; Greenblatt et al. 2024 ($\rightarrow$ Q28).